Vol 2 Chapter 12 — The Experimental Program: Substrate Engineering Falsifiers

Keeper (author pass)

2026-05-23 Saturday

Chapter 12 — The Experimental Program: Substrate Engineering Falsifiers

A framework is only as strong as its falsifiability. BST's experimental program — the SP-29 Casimir Mechanism Investigation and the SP-30 Substrate Engineering Program — collectively target the substrate's positive predictions for laboratory verification across a suite of experiments at 80K to 500K scales.

This chapter sets out the program's experimental targets, the substrate-mechanism predictions each test, and the per-experiment cost and timeline estimates. The program is what makes BST a falsifiable scientific framework rather than a theoretical exercise.

12.1 The substrate-engineering framework

The substrate is not directly probable at the Koons-tick scale (10^{-120} seconds) with current or foreseeable instruments. But the substrate's commitment-cycle output produces predictions at *observable* scales, and those predictions admit experimental tests.

The substrate-engineering framework — Casey's framing from May 2026 — is the methodology of designing experiments that probe substrate-derived predictions at scales where current technology can measure them. The SP-29 and SP-30 programs collectively target a suite of such experiments.

12.2 SP-29 Casimir Mechanism Investigation

The substrate's vacuum produces both the laboratory Casimir effect and the cosmological constant (Volume 4 Chapter 4 Λ-Casimir unification, Lyra T2418). SP-29 targets precision Casimir measurements that test substrate-specific corrections:

- H4 Cs-137 measurement: precision Casimir-force measurement using Cs-137 sources, testing the substrate's predicted Casimir corrections at the 0.1% level
- Cost: 60-90K
- Timeline: 6-12 months from procurement to first data
- Falsifier sharpness: HIGH (precision regime accessible)

12.3 SP-30 Substrate Engineering Program

The SP-30 program covers five experimental designs targeting different substrate predictions:

- SP-30-1 Precision Bell experiment (300-500K): tests the substrate sub-Tsirelson signature $|S|^2 \leq 126/16$ with the gap $1/8 = 1/2^{N_c}$ as the substrate-specific prediction. Outreach targets: Vienna, Caltech, Munich, Hanson.
- SP-30-2 Eigentone signal (200K): tests the substrate's predicted natural-frequency signatures via NIST/PTB/JILA/MPI atomic-clock infrastructure.

- SP-30-3 Commitment manipulation (80-150K): tests the substrate's predicted quantum-eraser revival amplitude correction at $1/N_{\max} \approx 0.73\%$ via precision delayed-choice quantum eraser. Filed paper-grade May 23, 2026.
- SP-30-4 Time-granularity measurement (200-400K): tests the substrate clock cycle $N_c \cdot t_{\text{Planck}}$ via next-generation optical lattice clocks at 10^{-19} precision.
- SP-30-5 Substrate parallelism architecture: theoretical-first development; experimental design follows.

Per-experiment timelines are 6-12 months from procurement; total program cost $\sim$ 640-900K.

12.4 The Five-Absence falsifiers

In parallel with the substrate-engineering positive-prediction tests, the Five-Absence framework (Chapter 11) provides experimental falsifiers via:

- Proton-decay searches (Super-Kamiokande, Hyper-Kamiokande)
- Magnetic-monopole searches (MoEDAL, IceCube)
- Sterile-neutrino searches (LSND/MiniBooNE/IceCube)
- Supersymmetric-particle searches (LHC, future colliders)
- Grand-unified-theory signatures (precision SM tests, proton decay)

All five experimental programs are currently running. BST predicts they will continue to find nothing positive; current results are consistent with this prediction.

12.5 The CPT falsifier

CPT violation at any precision would refute the substrate's $SO_0(5,2)$ structure (Volume 0 Chapter 8, structural CPT). Current experimental precision: $\sim 10^{-18}$ on proton/antiproton mass differences. BST predicts no CPT violation at any precision; the test will tighten as experimental precision improves.

12.6 What comes next

This is the final chapter of Volume 2. After Volume 2, the curriculum continues with:

- Volume 3 — Nuclear and Atomic Physics
- Volume 4 — General Relativity and Cosmology (Newton's G and the cosmological constant already rewritten as recruiter chapters)
- Volume 5 — Quantum Mechanics
- Volume 6 — Statistical Mechanics and Thermodynamics
- Volume 7 — Electromagnetism
- Volume 8 — Classical Mechanics
- Volume 9 — Condensed Matter Physics
- Volume 10 — Mathematical Methods
- Volume 11 — Generative Geometry and Topology
- Volume 12 — Chemistry
- Volume 13 — Biology
- Volume 14 — Information Theory
- Volume 15 — Methodology

The substrate framework continues to derive observables across every physics domain. Each volume applies the same substrate-mechanism pattern to the relevant physical regime.

Where to look this up: SP-29: `notes/SP29-1_H4_Cs137_paper_grade_proposal.md`. SP-30 series: per-sub-item paper-grade proposals (SP-30-1 through SP-30-5 filed). Five-Absence Set: `notes/Five_Absence_Predictions_Principle.md`.

For experimental status of the falsifiers: Particle Data Group annual reviews; relevant experiment collaboration publications.